# Software Engineering

## Assignment-9

## TYPE OF QUESTION: MCQ/MSQ

**Number of questions**: **10** **Total mark: 10 X 1 = 10**

For each of the following questions one or more of the given options are correct. Choose the correct options.

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**QUESTION 1:**

Which of the following are purposes of using CRC cards in object-oriented design?

a. Identifying class responsibilities
b. Discovering collaborations among classes
c. Replacing sequence diagrams with collaboration diagrams
d. Analysis of complex sequence diagrams
e. Determining low-level machine instructions

**Correct Answer: a.** Identifying class responsibilities , **b**. Discovering collaborations among classes

**Detailed Solution:**

CRC cards are used to identify class responsibilities and collaborations among classes during object-oriented design. They are not used for low-level implementation details.

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**QUESTION 2:**

Which of the following do NOT serve as useful guidelines for identifying boundary classes from a problem description?

a. Boundary classes often represent interfaces between the system and external actors
b. Every noun appearing in the problem description must be modeled as a boundary class
c. Screens, forms, and dialogs may correspond to boundary classes
d. Boundary classes are responsible for long-term data storage
e. External system interfaces may lead to identification of boundary classes

**Correct Answer: b.** Every noun appearing in the problem description must be modeled as a boundary class, **d.** Boundary classes are responsible for long-term data storage

**Detailed Solution:**

Not every noun becomes a boundary class, and boundary classes do not handle long-term data storage. They mainly represent interfaces between the system and external actors.

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**QUESTION 3:**

Which of the following statements represent software failures observed during testing?

a. The analyst omitted an important requirement from the specification
b. The programmer used '*' instead of '/' in an arithmetic expression
c. The application terminates unexpectedly for a valid user input
d. A tester observes an incorrect output for a valid transaction
e. A reviewer submits a report of a logical defect that was identified during code inspection
f. The software hangs when multiple users access it simultaneously

**Correct Answer: c.** The application terminates unexpectedly for a valid user input, **d.** A tester observes an incorrect output for a valid transaction, **f.** The software hangs when multiple users access it simultaneously

**Detailed Solution:**

A software failure is an observable incorrect behavior during execution, such as crashes, incorrect outputs, or system hangs.

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**QUESTION 4:**

Which of the following activities are primarily considered validation activities in software engineering?

a. Code inspection
b. Acceptance testing
c. System testing
d. User evaluation of prototypes
e. Design walkthrough

**Correct Answer: b.** Acceptance testing, **c**. System testing, **d**. User evaluation of prototypes

**Detailed Solution:**

Validation checks whether the software satisfies user needs. Acceptance testing, system testing, and user evaluation of prototypes are validation activities.

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**QUESTION 5:**

Which of the following are examples of failures rather than faults or errors in software engineering?

a. Incorrect operator used in source code
b. The software freezes during execution
c. A tester observes an unexpected output from the program
d. A programmer misunderstands the design specification
e. The system rejects a valid input value
f. A code inspection reveals a missing condition check

**Correct Answer: b**. The software freezes during execution, **c**. A tester observes an unexpected output from the program, **e**. The system rejects a valid input value

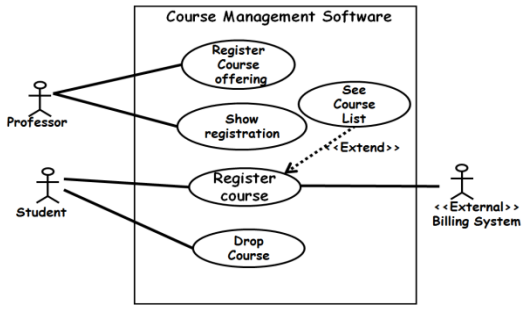
**Detailed Solution:**

Failures are observable incorrect behaviors during execution, such as freezing, wrong outputs, or rejecting valid inputs.

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**QUESTION 6:**

What is the number of boundary classes that must be included in the domain model for a software development project whose use case diagram is shown below?




a. 4
b. 5
c. 6
d. 7
e. 8

**Correct Answer: b.** 5

**Detailed Solution:**

The number of boundary classes depends on the actor–use case interactions shown in the diagram. So, number of boundary classes =5 .

**QUESTION 7:**

A software company observes that software testing consumes a large portion of total project effort, but testing activities are often completed within a relatively short calendar duration. Which of the following factors can explain this situation?

a. Multiple testing activities can be executed in parallel
b. Automated testing tools can significantly reduce testing time
c. Testing teams are usually larger than coding teams
d. Several testers can independently execute different test cases simultaneously
e. Testing is typically less complex than software design

**Correct Answer: a.** Multiple testing activities can be executed in parallel
**b.** Automated testing tolls can significantly reduce testing time
**d**. Several testers can independently execute different test cases simultaneously

**Detailed Solution:**

Testing can finish in shorter calendar time because many testing tasks and test cases can run in parallel, often supported by automation tools.

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**QUESTION 8:**

Which of the following statements are true regarding negative test cases?

a. Negative testing is mainly intended to confirm that the software crashes for incorrect inputs
b. Negative test cases verify whether the software can gracefully handle invalid inputs
c. Entering alphabetic characters into a numeric field is an example of negative testing
d. Negative test cases help improve software robustness and reliability
e. Negative testing is unnecessary if positive test cases have already passed

**Correct Answer: b.** Negative test cases verify whether the software can gracefully handle invalid inputs, **c**. Entering alphabetic characters into a numeric field is an example of negative testing, **d**. Negative test cases help improve software robustness and reliability

**Detailed Solution:**

Negative testing checks whether software handles invalid inputs properly. Invalid inputs like alphabets in numeric fields help improve robustness and reliability.

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**QUESTION 9:**

Which of the following statements are correct consequences of the pesticide effect in software testing?

a. The pesticide effect suggests that testing activities can stop once a test suite has been executed several times.
b. Repeating the same set of test cases indefinitely may result in fewer new defects being detected over time.
c. Test cases should be periodically reviewed, modified, and supplemented with new tests.
d. Executing the same test suite repeatedly guarantees that all remaining defects will eventually be found.
e. A test suite that was effective in the past may become less effective at finding new defects in the future.

**Correct Answer: b.** Repeating the same set of test cases indefinitely may result in fewer new defects being detected over time, **c.** Test cases should be periodically reviewed, modified, and supplemented with new tests , **e.** A test suite that was effective in the past may become less effective at finding new defects in the future

**Detailed Solution:**

The pesticide effect states that repeatedly running the same tests finds fewer new defects over time. Therefore, test cases should be updated regularly, as previously effective test suites may lose effectiveness.

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**QUESTION 10:**

Which of the following statements are true regarding who tests software?

a. Programmers commonly perform system testing
b. Users are mainly responsible for integration testing
c. Beta versions of software are often tested by a set of friendly customers
d. A dedicated test team may develop system test plans and testing strategies
e. Test teams usually perform all forms of testing except unit and acceptance testing

**Correct Answer: c.** Beta versions of software are often tested by a set of friendly customers
**d.** A dedicated test team may develop system test plans and testing strategies
**e**. Test teams usually perform all forms of testing except unit and acceptance testing

**Detailed Solution:**

Beta testing is often done by selected customers, while dedicated test teams handle most testing activities except unit testing and acceptance testing.

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